

1 Introduction

IEEE 802.11 wireless networks use a contention-based medium access mechanism to allow multiple stations to share a common wireless channel. The Distributed Coordination Function (DCF) uses Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) to coordinate channel access among competing stations.

In this laboratory, students will modify the MAC-layer channel access and backoff mechanism using the OMNeT++ simulation environment and the INET framework.

The objective is to understand how different backoff strategies affect wireless network performance.

The experiments will investigate the relationship between:

- contention and collision probability,
- backoff strategy and throughput,
- retransmission and packet delay,
- deterministic and random channel access,
- network size and channel utilization,
- fairness among competing wireless nodes.

Important: The experiments must use the same network topology, radio parameters, traffic parameters, and simulation duration when comparing different algorithms. Only the MAC channel-access/backoff algorithm should be changed.

2 Learning Objectives

1. Explain the channel-access mechanism of IEEE 802.11 DCF.
2. Explain the purpose of the contention window and backoff timer.
3. Implement alternative MAC backoff strategies.
4. Modify an existing INET MAC implementation.
5. Identify and measure packet collisions and retransmissions.
6. Evaluate wireless network throughput and delay.
7. Analyze the effect of the number of competing nodes.
8. Compare random and deterministic channel-access strategies.
9. Design an adaptive MAC backoff algorithm.

3 Background

3.1 IEEE 802.11 DCF

The Distributed Coordination Function is a fundamental channel-access mechanism used in IEEE 802.11 networks.

A simplified channel-access procedure is:

1. A station senses the wireless channel.
2. If the channel is busy, the station waits.
3. If the channel is idle, the station waits for the appropriate inter-frame spacing.
4. The station selects a backoff value.
5. The backoff timer decreases while the channel remains idle.
6. When the timer reaches zero, the station transmits.
7. If a collision occurs, the contention window is increased.
8. After a successful transmission, the contention window may be reset.

3.2 Contention Window

The contention window determines the range from which a station selects its random backoff.

If the current contention window is CW , the backoff can be represented as:

$$B = \text{Random}(0, CW)$$

where B represents the number of backoff slots.

The duration of the backoff is:

$$T_{\text{backoff}} = B \times T_{\text{slot}}$$

where T_{slot} is the duration of one backoff slot.

3.3 Binary Exponential Backoff

A common collision-response mechanism is binary exponential backoff. After a collision, the contention window is increased.

A simplified representation is:

$$CW_{new} = \min(2(CW_{old} + 1) - 1, CW_{max})$$

The objective is to reduce the probability that multiple stations select the same transmission opportunity.

3.4 INET Implementation

The lab uses the INET framework for OMNeT++. In an IEEE 802.11 network, the MAC layer contains the channel-access mechanism and associated backoff logic.

Students should inspect the relevant MAC implementation and identify the following concepts:

- channel sensing,
- backoff generation,
- backoff timer,
- contention window,
- collision/retransmission,
- successful transmission,
- retry limit.

4 Common Simulation Setup

All tasks should use a common simulation environment.

4.1 Recommended Parameters

The following parameters may be used as the baseline configuration.

Table 1: Common simulation parameters

Parameter	Value
Simulation Framework	OMNeT++ + INET 4.5
MAC	IEEE 802.11 DCF
Number of wireless nodes	3 initially
Number of wired nodes	3
Bitrate	2 Mbps
Transmission power	20 mW
Receiver sensitivity	−85 dBm
SNIR threshold	4 dB
Bandwidth	22 MHz
Background noise	−110 dBm
Packet size	100 B
Packet interval	1 s
Simulation time	2 s
Initial CW	15
Maximum CW	1023
Short retry limit	7

Students should increase the number of wireless nodes for scalability experiments.

Recommended node counts are:

$$N \in \{3, 5, 10, 20, 50, 100\}.$$

4.2 Important Experimental Rule

When comparing two algorithms, all parameters except the MAC algorithm must remain unchanged.

For example, the following should remain identical:

- topology,
- simulation time,
- random seed policy,
- packet size,
- packet generation rate,

- radio parameters,
- transmission power,
- receiver sensitivity,
- bandwidth.

5 Task 1 – Common Backoff with Collision-Based Doubling

5.1 Objective

Implement a channel-access mechanism where all nodes initially use the same backoff value.

The initial backoff value should be derived from the current contention-window configuration.

5.2 Requirements

Students must:

1. Assign the same initial backoff value to all competing nodes.
2. Use the current contention window to determine the initial value.
3. Detect a collision.
4. Double the backoff value after every collision.
5. Continue doubling until successful transmission or retry limit.
6. Reset the backoff to the initial value after successful transmission of a new packet.

For example, if the initial backoff is 15 slots:

$$15 \rightarrow 30 \rightarrow 60 \rightarrow 120 \rightarrow 240 \rightarrow \dots$$

The backoff should not exceed the configured maximum.

5.3 Expected Behavior

Because all nodes start with the same backoff value, simultaneous transmission attempts are likely.

Students should observe:

- increased collision probability,
- increased retransmission,
- reduced throughput under high contention,
- increased packet delay.

5.4 Questions

1. Why do simultaneous nodes tend to collide?
2. How does doubling the backoff affect collision probability?
3. How does the number of nodes affect throughput?
4. What happens when the number of nodes becomes large?
5. Compare the algorithm with standard IEEE 802.11 backoff.

6 Task 2 – Incremental Deterministic Initial Backoff

6.1 Objective

Implement an initial deterministic backoff assignment where each node receives a different backoff value.

The initial backoff should follow:

$$B_i = i + 1$$

where i is the node index.

For example:

Table 2: Example deterministic initial backoff

Node	Initial Backoff (slots)
0	1
1	2
2	3
3	4
4	5

6.2 Collision Behavior

After a collision, the node doubles its current backoff:

$$B_{new} = 2B_{old}.$$

For example:

$$3 \rightarrow 6 \rightarrow 12 \rightarrow 24.$$

After a successful transmission, the node returns to its assigned initial deterministic value for the next new packet.

6.3 Questions

1. Does deterministic assignment reduce collisions?
2. Does this algorithm improve throughput?
3. Does it introduce unfairness?
4. What happens when the number of nodes increases?
5. What happens when nodes are dynamically added or removed?

6.4 Required Experiment

Run the simulation using:

$$N = 5, 10, 20, 50, 100.$$

Plot:

- Number of nodes vs. collision probability.
- Number of nodes vs. throughput.
- Number of nodes vs. average delay.

7 Task 3 – Persistent Random Backoff

7.1 Objective

Implement a persistent random backoff mechanism.

Each node selects a random initial backoff value once and uses this value as its base backoff for subsequent new packets.

7.2 Initial Backoff

For each node:

$$B_i = \text{Random}(1, CW).$$

For example:

Table 3: Example random initial backoff

Node	Selected Backoff
0	7
1	12
2	3
3	19

The selected value becomes the node's persistent base backoff.

7.3 Collision Behavior

After collision:

$$B \rightarrow 2B \rightarrow 4B \rightarrow 8B \rightarrow \dots$$

until successful transmission or retry limit.

After successful transmission, the node returns to its original selected backoff.

For example:

$$7 \rightarrow 14 \rightarrow 28 \rightarrow \text{Success}$$

The next packet begins again with:

7.

7.4 Important Requirement

A new random base backoff must **not** be generated after every successful packet.

The node should retain its originally selected base value.

7.5 Questions

1. How does persistent randomization affect collisions?
2. What happens if two nodes select the same initial backoff?
3. How does increasing CW affect collision probability?
4. Is the algorithm fair among competing nodes?
5. How does it compare with Tasks 1 and 2?

8 Task 4 – Deterministic Backoff with Collision Randomization

8.1 Objective

Implement a hybrid channel-access mechanism that combines deterministic backoff with random recovery after collisions.

8.2 Required Behavior

The following state sequence should be implemented:

New Packet $\rightarrow$ Random Backoff $\rightarrow$ Successful Transmission $\rightarrow$ Deterministic Backoff

After successful deterministic transmissions, the node continues using the deterministic backoff.

If a collision occurs:

Deterministic Backoff $\rightarrow$ Collision $\rightarrow$ Random Backoff.

After successful recovery, the node returns to deterministic operation.

8.3 Deterministic Backoff Options

Students may use one of the following approaches.

8.3.1 Option A: Fixed Value Based

$$B_{det} = 7 \text{ or } 15 \dots$$

8.3.2 Option B: Previous successful backoff

The deterministic backoff is equal to the backoff value that resulted in the previous successful transmission.

Students must clearly state which method they use.

8.4 Questions

1. Can deterministic backoff reduce unnecessary randomness?
2. What happens when two nodes have the same deterministic backoff?
3. Does randomization after collision improve recovery?
4. Does this algorithm improve channel utilization?
5. Which algorithm provides the best fairness?

9 Task 5 – Advanced: Adaptive Contention Window

9.1 Objective

Design and implement an adaptive MAC backoff algorithm that dynamically adjusts the contention window according to observed network conditions.

Unlike the previous tasks, the algorithm should make decisions based on measured network behavior.

9.2 Network Statistics

Each node should maintain one or more of the following:

- number of transmission attempts,
- number of successful transmissions,
- number of collisions,

- number of consecutive collisions,
- recent channel-busy time,
- recent channel-idle time.

9.3 Collision-Based Adaptation

A simple adaptive strategy is:

$$CW_{new} = \begin{cases} \min(2CW_{old}, CW_{max}), & \text{if collision rate is high,} \\ \max(CW_{old}/2, CW_{min}), & \text{if collision rate is low.} \end{cases}$$

For example:

$$15 \rightarrow 30 \rightarrow 60 \rightarrow 120$$

under high contention.

When contention decreases:

$$120 \rightarrow 60 \rightarrow 30 \rightarrow 15.$$

9.4 Collision Rate

Students may calculate a recent collision rate using:

$$R_c = \frac{N_c}{N_t}$$

where:

- N_c is the number of collisions,
- N_t is the number of recent transmission attempts.

For example, if the recent history contains:

$$[1, 1, 0, 1, 0, 0, 0, 0],$$

then:

$$R_c = \frac{3}{8} = 0.375.$$

Students should define an appropriate threshold for increasing or decreasing the contention window.

9.5 Required Experiments

Evaluate the adaptive algorithm under:

1. low network load,
2. medium network load,
3. high network load,
4. increasing number of nodes,
5. bursty traffic,
6. saturated traffic.

9.6 Required Graphs

Students must produce:

- Number of nodes vs. throughput.
- Number of nodes vs. collision rate.
- Number of nodes vs. average delay.
- Number of nodes vs. average contention window.
- Traffic load vs. throughput.
- Traffic load vs. collision rate.

10 Performance Metrics

Students must collect and report the following metrics.

10.1 Throughput

Throughput can be calculated as:

$$Throughput = \frac{\text{Successfully delivered bits}}{\text{Simulation time}}.$$

The result should be reported in bits per second or Mbps.

10.2 Packet Delivery Ratio

$$PDR = \frac{N_{\text{received}}}{N_{\text{generated}}} \times 100\%.$$

10.3 Collision Probability

$$P_{\text{collision}} = \frac{N_{\text{collision}}}{N_{\text{attempt}}}.$$

10.4 Retransmission Count

Students should record the total number of retransmission attempts.

11 Comparison of Algorithms

Students must summarize the results of all tasks.

Table 4: Comparison of MAC backoff algorithms

Algorithm	Collision	Throughput	PDR	Retransmission Count
Task 1				
Task 2				
Task 3				
Task 4				
Task 5				

Students should provide a detailed explanation of the observed differences.

12 Required Graphs

At minimum, the report should contain the following graphs:

1. Number of nodes vs. throughput.
2. Number of nodes vs. collision probability.

3. Number of nodes vs. retransmission count.

For Task 5, additionally include:

6. Traffic load vs. throughput.
7. Traffic load vs. collision probability.
8. Traffic load vs. average contention window.

13 Experimental Procedure

For each task:

1. Implement the required MAC algorithm.
2. Compile the modified INET framework.
3. Verify that the simulation runs without errors.
4. Run the simulation using the baseline topology.
5. Repeat the experiment with different numbers of nodes.
6. Collect scalar and vector statistics where appropriate.
7. Calculate the required performance metrics.
8. Generate graphs.
9. Compare the results with the baseline IEEE 802.11 mechanism.

Students should use multiple simulation runs when statistical confidence is required.

14 Lab Report Requirements

The final report should contain:

1. Introduction
2. Objectives
3. Network topology
4. Simulation parameters
5. Explanation of the original IEEE 802.11 backoff mechanism
6. Implementation methodology for each task
7. Important code modifications

8. Experimental results
9. Graphs
10. Comparison of algorithms
11. Discussion
12. Answers to task-specific questions
13. Conclusion

Students should include only the relevant portions of modified source code rather than submitting entire INET source files.